Probability

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1 Introduction

Definition 1.1. *(Probability)*
A probability P on a sample space Ω satisfies

- $P(A) \geq 0$;
- $P(\Omega) = 1$;
- if $A \cap B = \varnothing$, then $P(A \cup B) = P(A) + P(B)$.

Example.
Buffon's needle. A needle of length l is dropped on a floor ruled with parallel lines a distance a apart, where $a > l$. Let $\theta \in [0, \pi]$ be the angle between the needle and the direction of the lines, and let $y \in [0, a/2]$ be the distance from the center of

the needle to the nearest line. The sample space and the event that the needle crosses a line are

$$\Omega = \{(\theta, y) : 0 \le \theta \le \pi, 0 \le y \le a/2\},$$

$$A = \{(\theta, y) : 0 \le \theta \le \pi, 0 \le y \le (l/2)\sin\theta\}.$$

Comparing areas,

$$P(A) = \frac{\int_0^\pi (l/2)\sin\theta \, \mathrm{d}\theta}{\pi a/2} = \frac{2l}{\pi a}.$$

Definition 1.2. *(Conditional Probability)*
If $P(B) > 0$, then $P(A|B) = \frac{P(A \cap B)}{P(B)}$.

Definition 1.3. *(Independence)*
Events A and B are independent if $P(A \cap B) = P(A)P(B)$.

Remark.
Mutual independence of k events does not imply mutual independence of $k+1$ events.

Theorem 1.1. *(Total Probability)*
If the events B_k are pairwise disjoint and $A \subset \bigcup_k B_k$, then

$$P(A) = \sum_k P(B_k)P(A|B_k).$$

Theorem 1.2. *(Bayes)*
Under the same hypotheses, and if $P(A) > 0$,

$$P(B_k|A) = \frac{P(B_k)P(A|B_k)}{\sum_j P(B_j)P(A|B_j)}.$$

Theorem 1.3. *(Borel-Cantelli)*
Let A_n be events and write A_n i.o. for the event that infinitely many of them occur, so that $P(A_n \text{ i.o.}) = P(\limsup A_n)$.

- If $\sum_{n=1}^{\infty} P(A_n) < \infty$, then $P(A_n \text{ i.o.}) = 0$.
- If $\sum_{n=1}^{\infty} P(A_n) = \infty$ and the events A_n are mutually independent, then $P(A_n \text{ i.o.}) = 1$.

2 Random Variables

2.1 Distribution Functions

Definition 2.1. *(Distribution Function)*
The distribution function of a random variable X is $F(x) = P(X \leq x)$. It is nondecreasing and right-continuous, its left limits exist, and $F(-\infty) = 0$, $F(+\infty) = 1$.

Definition 2.2. *(Density)*
If $F(x) = \int_{-\infty}^{x} f(u)\,\mathrm{d}u$ with f a nonnegative finite function, then f is a density of X and X is a continuous random variable. In that case
- $\int_{-\infty}^{+\infty} f(u)\,\mathrm{d}u = 1$;
- if f is continuous at x_0, then $F'(x_0) = f(x_0)$;
- if $B \in \mathcal{B}(\mathbb{R})$, then $P(X \in B) = \int_B f(x)\,\mathrm{d}x$.

Proposition 2.1.
If $Y = g(X)$, then $P(Y = y) = P(X \in g^{-1}(\{y\}))$ and $F_Y(y) = P(g(X) \leq y)$. If g is strictly monotone and continuously invertible, then

$$f_Y(y) = f_X(g^{-1}(y))\left|(g^{-1})'(y)\right|.$$

2.2 Expectation and Variance

Definition 2.3. *(Expectation)*
If X is discrete with values x_k, then $E(X) = \sum_k x_k P(X = x_k)$. If X is continuous with density f, then $E(X) = \int_{-\infty}^{+\infty} x f(x)\,\mathrm{d}x$. For a general random variable, $E(g(X)) = \int_{-\infty}^{+\infty} g(x)\,\mathrm{d}F_X(x)$.

Remark.
- If X is nonnegative and integer-valued, then $E(X) = \sum_{k=0}^{\infty} P(X > k)$.
- If X is nonnegative, then $E(X) = \int_0^{\infty} P(X > t)\,\mathrm{d}t$.
- The expectation need not exist.

Definition 2.4. *(Variance)*

$$\mathrm{Var}(X) = E(X - E(X))^2 = E(X^2) - (E(X))^2.$$

The handwritten notes denote this by $D(X)$.

Proposition 2.2.

- For every constant x_0, $\mathrm{Var}(X) \le E(X - x_0)^2$.
- $\mathrm{Var}(X) = 0$ if and only if X is almost surely constant.

Theorem 2.1. *(Chebyshev)*
If $\mathrm{Var}(X) < \infty$, then for every $\varepsilon > 0$,

$$P(|X - E(X)| \ge \varepsilon) \le \frac{\mathrm{Var}(X)}{\varepsilon^2}.$$

2.3 Common Distributions

Definition 2.5. *(Bernoulli and Binomial)*
A Bernoulli random variable with success probability p satisfies $P(X = k) = p^k(1-p)^{1-k}$ for $k = 0, 1$. Its mean is p and its variance is $p(1-p)$.
A binomial random variable, the number of successes in n independent trials, satisfies

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}.$$

Its mean is np and its variance is $np(1-p)$.

Definition 2.6. *(Hypergeometric)*
From N balls of which K are white, draw n balls without replacement, and let X be the number of white balls drawn. Then

$$P(X = k) = \frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}},$$

$$E(X) = nK/N, \quad \mathrm{Var}(X) = n(K/N)(1 - K/N)\frac{N-n}{N-1}.$$

Definition 2.7. *(Geometric and Negative Binomial)*
The number of trials until the first success, with success probability p, satisfies $P(X = k) = (1-p)^{k-1}p$ for $k = 1, 2, \ldots$. Its mean is $1/p$ and its variance is $(1-p)/p^2$.
The number of trials until the r-th success satisfies

$$P(X = k) = \binom{k-1}{r-1}(1-p)^{k-r}p^r, \quad k = r, r+1, \ldots.$$

Its mean is r/p and its variance is $r(1-p)/p^2$.

Definition 2.8. *(Poisson)*

$$P(X = k) = e^{-\lambda}\frac{\lambda^k}{k!}$$

for $k = 0, 1, 2, \ldots$. The mean and the variance are both λ. This is the law of the number of events in a fixed interval.

Definition 2.9. *(Discrete Power Law)*

$$P(X = k) = c/k^{\alpha+1},$$

where c normalizes the probabilities. The mean and the variance need not be finite, since the mass is highly uneven.

Definition 2.10. *(Uniform, Pareto, and Cauchy)*
The uniform law on (a, b) has density $1/(b-a)$, mean $(a+b)/2$, and variance $(b-a)^2/12$.
The Pareto density $\alpha/x^{\alpha+1}$ on $x \geq 1$, with $\alpha > 0$, has mean $\alpha/(\alpha-1)$ for $\alpha > 1$ and variance $\alpha/((\alpha-1)^2(\alpha-2))$ for $\alpha > 2$.
The Cauchy density $1/(\pi(1+x^2))$ has neither a mean nor a variance.

Definition 2.11. *(Exponential and Gamma)*
The exponential density $\lambda e^{-\lambda x}$ for $x \geq 0$ and $\lambda > 0$ has mean $1/\lambda$ and variance $1/\lambda^2$.
The gamma density, with shape $\alpha > 0$ and rate $\lambda > 0$, is

$$f(x) = (\lambda^\alpha/\Gamma(\alpha))e^{-\lambda x}x^{\alpha-1}, \quad x \geq 0.$$

Its mean is α/λ and its variance is α/λ^2. Here $\Gamma(\alpha) = \int_0^\infty t^{\alpha-1}e^{-t}\,\mathrm{d}t$, so $\Gamma(n+1) = n\Gamma(n)$ and $\Gamma(n) = (n-1)!$ for positive integers n.

Definition 2.12. *(Normal and Lognormal)*
The normal density with mean μ and variance $\sigma^2 > 0$ is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

The standard density is $\varphi(x) = 1/\sqrt{2\pi}e^{-x^2/2}$, with mean 0 and variance 1. If $X \sim N(\mu, \sigma^2)$, then $(X-\mu)/\sigma \sim N(0,1)$. The distribution function Φ of $N(0,1)$ satisfies $\Phi(-x) = 1 - \Phi(x)$.
If $\log Y \sim N(\mu, \sigma^2)$ with $Y > 0$, then

$$f_Y(y) = \frac{1}{y\sigma\sqrt{2\pi}} \exp\left(-\frac{(\log y - \mu)^2}{2\sigma^2}\right).$$

The mean is $e^{\mu+\sigma^2/2}$ and the variance is $\left(e^{\sigma^2}-1\right)e^{2\mu+\sigma^2}$.

3 Random Vectors

3.1 Joint Distributions

Definition 3.1. *(Joint Distribution)*

$$F(x_1, ..., x_n) = P(X_1 \le x_1, ..., X_n \le x_n).$$

The joint distribution function is 0 when any coordinate tends to $-\infty$, and it is 1 when every coordinate tends to $+\infty$. Every marginal distribution is obtained by sending the remaining coordinates to $+\infty$.

Proposition 3.1.
$X_1, ..., X_n$ are independent if and only if $F(x_1, ..., x_n) = F_{X_1}(x_1)...F_{X_n}(x_n)$. The same factorization holds for densities.

Definition 3.2. *(Joint Density)*
When the mixed partial exists,

$$f(x_1, ..., x_n) = \frac{\partial^n F}{\partial x_1 ... \partial x_n},$$

and the marginal density of $(X_1, ..., X_k)$ is obtained by integrating out $x_{k+1}, ..., x_n$.

Definition 3.3. *(Conditional Distribution)*

$$F(y|x) = \frac{\int_{-\infty}^{y} f(x,t)\,\mathrm{d}t}{f(x)}, \quad f(y|x) = \frac{f(x,y)}{f(x)}.$$

Remark.
The binomial family with a fixed success probability, the Poisson family, and the normal family are closed under addition of independent random variables.

Proposition 3.2.
If $y = y(x)$ is a smooth invertible change of variables with Jacobian $J = \frac{\partial(x_1,...,x_n)}{\partial(y_1,...,y_n)}$, then

$$f_Y(y_1, ..., y_n) = f_X(x_1, ..., x_n)|J|.$$

Definition 3.4. *(Order Statistics)*
The order statistics $X_{(1)} \leq ... \leq X_{(n)}$ are the sample arranged in increasing order.

Proposition 3.3.
If $X_1, ..., X_n$ are i.i.d. with distribution function F and density f, then

$$F_{X_{(n)}}(x) = F(x)^n, \quad F_{X_{(1)}}(x) = 1 - (1 - F(x))^n,$$

and the density of the k-th order statistic is

$$f_{X_{(k)}}(x) = \frac{n!}{(k-1)!(n-k)!} F(x)^{k-1}(1 - F(x))^{n-k} f(x).$$

On the region $x_1 < x_2 < ... < x_n$, the joint density of the order statistics is $n! f(x_1)...f(x_n)$.

3.2 Covariance and Conditional Expectation

Proposition 3.4.
Expectation is linear: $E\left(\sum_k c_k X_k\right) = \sum_k c_k E(X_k)$. If the summands are uncorrelated, then $\operatorname{Var}\left(\sum_k c_k X_k\right) = \sum_k c_k^2 \operatorname{Var}(X_k)$.

Definition 3.5. *(Covariance and Correlation)*

$$\operatorname{Cov}(X,Y) = E((X-EX)(Y-EY)) = E(XY) - E(X)E(Y).$$

The sign of the covariance is the sign of the correlation: positive, zero, or negative, in which case one says positively correlated, uncorrelated, or negatively correlated. The correlation is the normalized covariance,

$$\rho(X,Y) = \frac{\operatorname{Cov}(X,Y)}{\sqrt{\operatorname{Var}(X)\operatorname{Var}(Y)}}.$$

Proposition 3.5.

- Independence implies that the covariance is zero. The converse is false.
- If $ab > 0$, then $\rho(aX, bY) = \rho(X,Y)$.
- $\operatorname{Var}\left(\sum_k c_k X_k\right) = \sum_i \sum_j c_i c_j \operatorname{Cov}\left(X_i, X_j\right)$.
- $|\rho(X,Y)| \le 1$. Equality to 1 holds if and only if $Y = (\sigma_Y/\sigma_X)(X-EX) + EY$, and equality to -1 holds if and only if the same identity holds with the opposite sign in front of the slope.

Definition 3.6. *(Covariance Matrix)*
The matrix Σ with entries $\Sigma_{ij} = \operatorname{Cov}\left(X_i, X_j\right)$ is symmetric and positive semidefinite.

Definition 3.7. *(Conditional Expectation and Variance)*

$$\operatorname{Var}(Y|X=x) = E(Y^2|X=x) - (E(Y|X=x))^2.$$

Proposition 3.6.

- $E(Y) = E(E(Y|X))$.
- $E(g(X)Y \mid X) = g(X)E(Y|X)$.
- $E(Y - E(Y|X))^2 \le E(Y-g(X))^2$ for every function g.

Theorem 3.1. *(Law of Total Variance)*

$$\operatorname{Var}(Y) = E(\operatorname{Var}(Y|X)) + \operatorname{Var}(E(Y|X)).$$

Proof.

$$\operatorname{Var}(Y) = E(Y^2) - (EY)^2 = E(E(Y^2|X)) - (E(E(Y|X)))^2.$$

Substitute $E(Y^2|X) = \operatorname{Var}(Y|X) + (E(Y|X))^2$. The cross term that remains is $\operatorname{Var}(E(Y|X))$. ■

3.3 Multivariate Distributions

Definition 3.8. *(Multinomial)*
In n independent trials with r outcomes of probabilities $p_1, \ldots, p_r$, let X_j be the number of times outcome j occurs. If $k_1 + \ldots + k_r = n$, then

$$P(X_1 = k_1, \ldots, X_r = k_r) = \frac{n!}{k_{1!} \ldots k_{r!}} p_1^{k_1} \ldots p_r^{k_r}.$$

The same probability is the product of successive binomial coefficients $\binom{n}{k_1}\binom{n-k_1}{k_2}\ldots$.

Example.
For n independent shots at a target whose rings are labeled $0, \ldots, 10$, the vector of ring counts is multinomial.

Definition 3.9. *(Multivariate Hypergeometric)*
A population of N balls contains N_j balls of color j, with $\sum_j N_j = N$. Draw n balls without replacement, and let n_j be the number drawn of color j. Then

$$P(n_1, \ldots, n_r) = \frac{\prod_j \binom{N_j}{n_j}}{\binom{N}{n}}.$$

Definition 3.10. *(Uniform Distribution on a Region)*
On a region D of content $S(D)$, the uniform density is $1/S(D)$.

Definition 3.11. *(Bivariate Normal)*
The law $N(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$ has density

$$f(x_1, x_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left(-\frac{Q}{2(1-\rho^2)}\right),$$

where

$$Q = \frac{(x_1-\mu_1)^2}{\sigma_1^2} + \frac{(x_2-\mu_2)^2}{\sigma_2^2} - 2\rho(x_1-\mu_1)\frac{x_2-\mu_2}{\sigma_1\sigma_2}.$$

4 Generating Functions

4.1 Probability Generating Functions

Definition 4.1. *(Probability Generating Function)*
If X is nonnegative and integer-valued, then $G_X(s) = E(s^X) = \sum_k s^k P(X = k)$.

Proposition 4.1.
$G_X(1) = 1$ and $G_{X'}(1) = E(X)$. The generating function determines the distribution, through $P(X = k) = G_X^{(k)}\frac{0}{k!}$.
If $X_1, ..., X_n$ are independent, then $G_{X_1+...+X_n}(s) = \prod_{k=1}^n G_{X_k}(s)$. If in addition the X_i are i.i.d. and independent of a nonnegative integer-valued random index N, then the sum of the first N of them has generating function $G_N(G_X(s))$.

4.2 Characteristic Functions

Definition 4.2. *(Characteristic Function)*

$$\psi_X(t) = E(e^{itX}) = E(\cos tX) + iE(\sin tX).$$

Proposition 4.2.
- $|\psi_X(t)| \le \psi_X(0) = 1$, and $\psi_X(t) = \overline{\psi_X(-t)}$.
- $\psi_X^{(k)}\frac{0}{i^k} = E\big(X^k\big)$ when the moment exists.
- If $X_1, ..., X_n$ are independent, then $\psi_{X_1+...+X_n}(t) = \prod_{k=1}^n \psi_{X_k}(t)$.
- The characteristic function determines the distribution. If it is integrable and X has a density, then

$$f(x) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} e^{-itx}\psi_X(t)\,\mathrm{d}t.$$

Remark.
Common characteristic functions:
- two-point law, with $q = 1 - p$: $pe^{it} + q$;
- binomial: $\left(pe^{it} + q\right)^n$;
- Poisson: $e^{\lambda(e^{it}-1)}$;
- geometric, as the number of trials until the first success: $p\frac{e^{it}}{1-qe^{it}}$;
- uniform on $[a, b]$: $\frac{e^{itb}-e^{ita}}{it(b-a)}$;
- exponential of rate λ: $\frac{\lambda}{\lambda-it}$;
- standard Cauchy: $e^{-|t|}$;
- $N(\mu, \sigma^2)$: $e^{i\mu t-\sigma^2 t^2/2}$, and in particular $e^{-t^2/2}$ for $N(0, 1)$.

5 Limit Theorems

5.1 Laws of Large Numbers

Definition 5.1. *(Convergence in Probability)*
One writes $X_n \xrightarrow{P} X$ if for every $\varepsilon > 0$,

$$\lim_{n\to\infty} P(|X_n - X| < \varepsilon) = 1,$$

or equivalently $P(|X_n - X| \geq \varepsilon) \to 0$.

Definition 5.2. *(Weak and Strong Laws)*
The sequence $\{X_n\}$ obeys the weak law of large numbers if $(1/n)\sum_{k=1}^{n} X_k \xrightarrow{P} \alpha$ for some constant α. It obeys the strong law if the same average converges almost surely.

Theorem 5.1.
If $\operatorname{Cov}(X_i, X_j) \leq 0$ for $i \neq j$ and $(1/n^2)\sum_{k=1}^{n} \operatorname{Var}(X_k) \to 0$, then the centered averages of $\{X_n\}$ converge to 0 in probability.

Theorem 5.2. *(Kolmogorov)*
Let X_k be independent with finite variances, and let $S_k = \sum_{j=1}^{k}(X_j - EX_j)$. Then for every $\varepsilon > 0$,

$$P\left(\max_{1\leq k\leq n} |S_k| \geq \varepsilon\right) \leq (1/\varepsilon^2)\sum_{k=1}^{n} \operatorname{Var}(X_k).$$

If in addition $\sum_{k=1}^{\infty} \operatorname{Var}(X_k)/k^2 < \infty$, then $(1/n)\sum_{k=1}^{n}(X_k - EX_k) \to 0$ almost surely.

Proposition 5.1.
Let X_i be i.i.d. If f is continuous, then $f\Big((1/n)\sum_{k=1}^{n} X_k\Big) \to f(E(X_1))$ almost surely whenever the strong law applies to $\{X_k\}$. If $E|h(X_1)| < \infty$, then $(1/n)\sum_{k=1}^{n} h(X_k) \to E(h(X_1))$ almost surely.

Remark.
An i.i.d. sequence with finite expectation obeys both the weak and the strong law of large numbers.

5.2 The Central Limit Theorem

Remark.
Let X_i be i.i.d. with mean μ and variance $\sigma^2 < \infty$, and set $Y_n = \frac{\sum_{k=1}^{n}(X_k - \mu)}{n^{\alpha}}$. Then $E(Y_n) \to 0$ and

$$\lim_{n\to\infty} \operatorname{Var}(Y_n) = \begin{cases} 0 & \text{if } \alpha > 1/2 \\ \sigma^2 & \text{if } \alpha = 1/2. \\ \infty & \text{if } \alpha < 1/2 \end{cases}$$

The critical scaling is therefore $n^{1/2}$.

Theorem 5.3. *(Central Limit Theorem)*
If X_i are i.i.d. with mean μ and variance $\sigma^2 \in (0, \infty)$, then

$$\lim_{n\to\infty} P\left(\frac{\sum_{k=1}^{n}(X_k - \mu)}{\sigma\sqrt{n}} \le x\right) = \Phi(x).$$

The standardized partial sums converge in distribution to $N(0,1)$.

Theorem 5.4. *(Delta Method)*
Let X_i be i.i.d. with mean μ and variance σ^2, and let g be continuously differentiable with $g'(\mu) \neq 0$. Then $g\big(\overline{X}_n\big)$ is approximately normal with mean $g(\mu)$ and variance $\sigma^2 (g'(\mu))^2 / n$.

Theorem 5.5. *(Lindeberg--Feller)*
Let X_k be independent, with mean μ_k and variance σ_k^2, and set $B_n^2 = \sum_{k=1}^n \sigma_k^2$. Then $\max_{k \le n} \sigma_k^2 / B_n^2 \to 0$ and

$$\frac{\sum_{k=1}^n (X_k - \mu_k)}{B_n} \xrightarrow{d} N(0,1)$$

if and only if the Lindeberg condition holds: for every $\varepsilon > 0$,

$$\lim_{n\to\infty} (1/B_n^2) \sum_{k=1}^n E\Big((X_k - \mu_k)^2 I_{\{|X_k - \mu_k| \ge \varepsilon B_n\}}\Big) = 0.$$

Theorem 5.6. *(Lyapunov)*
Let X_k be independent with means μ_k and with $B_n^2 = \sum_{k=1}^n \mathrm{Var}(X_k)$. If for some $\delta > 0$,

$$\frac{\sum_{k=1}^n E|X_k - \mu_k|^{2+\delta}}{B_n^{2+\delta}} \to 0,$$

then $\frac{\sum_{k=1}^n (X_k - \mu_k)}{B_n} \xrightarrow{d} N(0,1)$.

Theorem 5.7.
Let $X_k = \big(X_{k1}, \ldots, X_{kp}\big)$ be i.i.d. random vectors in $\mathbb{R}^p$ with mean μ and covariance matrix Σ. Then

$$\big(1/\sqrt{n}\big) \sum_{k=1}^n (X_k - \mu) \xrightarrow{d} N(0, \Sigma).$$

Remark.
The limit theorems in this section require a finite variance.

5.3 Modes of Convergence

Definition 5.3. *(Four Modes)*
Write $X_n \xrightarrow{\text{a.s.}} X$ for almost sure convergence, $X_n \xrightarrow{P} X$ for convergence in probability, and $X_n \xrightarrow{d} X$ for convergence in distribution. Write $X_n \xrightarrow{L^r} X$ if $E|X_n - X|^r \to 0$. The last mode requires $E|X_n|^r < \infty$ and $E|X|^r < \infty$.

Proposition 5.2.

- If $X_n \xrightarrow{\text{a.s.}} X$ and $Y_n \xrightarrow{\text{a.s.}} Y$, then $X_n \pm Y_n \xrightarrow{\text{a.s.}} X \pm Y$ and $X_n Y_n \xrightarrow{\text{a.s.}} XY$. If also $Y_n \neq 0$ and $Y \neq 0$ almost surely, then $X_n / Y_n \xrightarrow{\text{a.s.}} X/Y$.
- The same three rules hold for convergence in probability.
- If $X_n \xrightarrow{d} X$, $Y_n \xrightarrow{d} a$, and $b_n \to b \neq 0$, then $b_n X_n \pm Y_n \xrightarrow{d} bX \pm a$. If $a = 0$, then $X_n Y_n \xrightarrow{P} 0$. If $a \neq 0$, then $X_n Y_n \xrightarrow{d} aX$ and $X_n / Y_n \xrightarrow{d} X/a$.
- Sums preserve L^r convergence. If $r = 2$, the product converges in L^1. If $|X_n|$ and $|Y_n|$ are essentially bounded by the same constant, the product converges in L^r. If $|X_n| \leq M$ and $|Y_n| \geq A > 0$ almost surely, the quotient converges in L^r.

Remark.

Almost sure convergence and L^r convergence both imply convergence in probability, which implies convergence in distribution.